
D-ITG :: Distributed Internet Traffic Generator
Reference Manual
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www.grid.unina.it/software/ITG

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1 - ITGSend

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1.1 Synopsis:

In case of using a script file to generate multiple flows type:
./ITGSend <script_file> [-m <msr_type>] [-l <logfile>]

See Section 1.7 for details

If you want to remotely control the sender, launch it in daemon mode (see Section 4 for details):
./ITGSend -Q [-m <msr_type>] [-l <logfile>]

Otherwise:

```
./ITGSend [-m <msr_type>] [-a <destination_address>] [-p <destination_port>] [-T
<protocol_type>] [-f <TTL>] [-b <DS byte>] [-s <seed>] [-L <log-server_addr>
<log_server_port> <protocol_type>] [-t <duration>] [-d <gen_delay>] [-l <logfile>]
[[-C <pkts_per_sec> | -U <min_pkts_per_sec max_pkts_per_sec> | -E
<average_pkts_per_sec> | -V <shape scale> | -Y <shape scale> | -N <mean std_dev> |
-O <average_pkts_per_sec> | -G <shape scale>]
[-c <pkt_size> | -u <min_pkt_size max_pkt_size> | -e <average_pkt_size> | -v <shape
scale> | -y <shape scale> | -n <mean std_dev> | -o <average_pkt_size> | -g <shape
scale>]] |
[[Telnet]][DNS]][[VoIP] [-x <codec_type>] [-h <protocol_type> -i
<Voice_Activity_Detection>]]]
```

1.2 Options:

-m	type of meter See 1.3 for details	
-a	destination_address DEFAULT: localhost	
-p	destination_port DEFAULT: 8999	
-T	protocol_type VALUES: UDP, TCP, ICMP DEFAULT: UDP	
If the used protocol is ICMP you must indicate the type of message. Over Linux platform the root privileges are needed.		
-b	DS byte DEFAULT: 0	Set the DS byte for QoS. It's expressed in decimal notation. DS $\in$ [0, 255].
-s	seed DEFAULT: Random	Set the seed for random number generator
-t	duration DEFAULT: 10000 msec	Set the generation duration. It's expressed in msec.
-d	gen_delay DEFAULT: 0 msec	Set the generation delay. It's expressed in msec.
-f	TTL DEFAULT: 64	Set time to live (TTL). It's expressed in decimal notation. TTL $\in$ [0, 255].
-l	logfile See 1.4 for details.	Generate log file.
-L	remote log See 1.5 for details	

Inter-departure time options:

-C	pkts_per_sec	-Constant inter-departure with specified packet rate.
-U	min_pkts_per_sec max_pkts_per_sec	-Uniformly distributed inter-departure
-E	average_pkts_per_sec	-Exponentially distributed inter-departure
-V	shape scale	-Pareto distributed inter-departure
-Y	shape scale	-Cauchy distributed inter-departure
-N	mean std_dev	-Normal distributed inter-departure
-O	average_pkts_per_sec	-Poisson distributed inter-departure
-G	shape scale	-Gamma distributed inter-departure

NOTE: If you don't specify any inter_departure time option the default behaviour will be: Constant inter-departure with 75000 packets per second.

Packet size options:

-c	pkt_size	-Constant payload size
-u	min_pkt_size max_pkt_size	-Uniformly distributed payload size
-e	average_pkt_size	-Exponentially distributed payload size
-v	shape scale	-Pareto distributed payload size
-y	shape scale	-Cauchy distributed payload size
-n	mean std_dev	-Normal distributed payload size
-o	average_pkt_size	-Poisson distributed payload size
-g	shape scale	-Gamma distributed payload size

NOTE: If you don't specify any packet size option the default behaviour will be: Constant payload size c = 1024 bytes

Application Level protocol indication:

Telnet	Generate traffic with Telnet traffic characteristics. It doesn't need any other option. NOTE: Telnet traffic generation works with (i) OWDMeter
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and (ii) TCP transport layer protocol. Different settings will be ignored.

VoIP

Generate traffic with VoIP traffic characteristics. See 1.6 for further details.

NOTE: VoIP traffic generation works with (i) OWDMeter and (ii) UDP transport layer protocol. Different settings will be ignored.

DNS

Generate traffic with DNS traffic characteristics. It doesn't need any other option.

NOTE: DNS traffic generation works with (i) OWDMeter using (ii) both UDP and TCP transport layer protocol. Different settings will be ignored.

NOTE: If you specify any application level protocol then you can't specify any inter-departure time options and any packet size options; if you want to specify any application level protocol you must indicate it after option. Other options indicated in Section 1.2 are applicable.

1.3 Type of meter:

VALUES: owdm (one way delay meter)
rttm (round trip time meter)

DEFAULT: owdm

In the case of rttm the sender log file is mandatory. You can specify the log file name using the option -l. The default name is ITGSend.log.

1.4 Log File:

log_file generates the log file

If the type of meter is OWDM, by omitting this option you will not generate log file. Otherwise, if the type of meter is RTTM, by omitting this option the program will generate a default log file as indicated below.

DEFAULT: /tmp/ITGSend.log

1.5 Remote Log :

< log_address> server-log IP address
DEFAULT: localhost

< log_port> server-log PORT
DEFAULT: 9002

<protocol_type> protocol used for communication between sender and log-server
VALUES: UDP, TCP
DEFAULT: UDP

1.6 Application Level Protocols Options:

VoIP options:

-x codec_type Set the Codec type
VALUES: G.711.1 for G.711 codec with 1 sample per pkt
G.711.2 for G.711 codec with 2 samples per pkt
G.723.1 for G.723.1 codec
G.729.2 for G.729 codec with 2 samples per pkt
G.729.3 for G.729 codec with 3 samples per pkt

DEFAULT: G.711 with 1 sample per pkt

-h protocol_type Set the protocol type
VALUES: RTP for Real Time Protocol
CRTP for Real Time Protocol with header compression

DEFAULT: RTP

-i Voice_Activity_Detection Set the Voice Activity Detection
VALUES: VAD for Voice Activity Detection on
DEFAULT: Voice Activity Detection off

In the case of VoIP traffic generation, the indication of the options presented in Section 1.2 can not follow the indication of the VoIP protocol. Digging into more details, a command line must be the following:

`./ITGSend [0 or more options present in the Section 1.2] VoIP [0 or more VoIP options]`

In the case of a number of options equal to “0” the generation starts with default parameters (see example 1).

Examples:

- 1) `./ITGSend VoIP`
- 2) `./ITGSend -l logfilename -t 30000 -d 5000 VoIP -x G.711.2 -h CRTP -i VAD`
- 3) `./ITGSend -l logfilename -t 30000 -p 6666 -b 128 VoIP`

1.7 Script file:

All generate flows must have the same type of meter (rttm or owdm).

<protocol_type> protocol used for communication between receiver and log-server
VALUES: UDP, TCP
DEFAULT: UDP

3 - ITGLog

3.1 Synopsis:

`./ITGLog [-l <local_logfile>]`

3.2 Options:

`-l` `logfile` Generate log file.
 DEFAULT: `/tmp/ITGLog.log`

4 - ITGapi

ITGapi is a C++ API that enables to remotely control traffic generation. For this purpose, after having launched ITGSend in daemon mode (./ITGSend -Q) on one or more traffic source nodes, it is possible to use the following function to remotely coordinate the traffic generation from the different senders:

```
int DITGsend(char *sender, char *message);
```

sender is the IP address of ITGSend and message is the string you would type at command line (except the name of the ITGSend executable file).

This prototype is declared in ITGapi.h

5 – WB::D-ITG: Web Based Distributed Internet Traffic Generator

We are working on a Web Based version of D-ITG. Currently we are testing it before the official releasing. It will be soon available at D-ITG web site.

6 – Results Analysis

D-ITG provides different tools for results analysis. For Windows and Linux Familiar platform there is the wdecod.exe utility. For Linux platforms there are the utility decod and mcal.
Digging into more details:

Windows and Linux Familiar platforms

wdecod.exe Binary_Log_file_name [-v] [-i] [-l outlog] [-d DT] [-j JT] [-b BT]:

-v -> generate total result for visualization
-i -> generate total result for import
-l outlog -> generate decoded log file (in text version) with name 'outlog'
-d DT -> generate delay file with average delay every DT millisec
-j JT -> generate jitter file with average jitter every JT millisec
-b BT -> generate bit rate file with average bitrate every BT millisec

Output files can be plotted with other utility.

Linux platforms

1 - decod converts the binary file in the text file.
2 – mcalc analyzes the text file

decod.exe Binary_Log_file_name Text_Log_file_name

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